forms the same office for the sensory nerves; in fact, whatever powers originate in the brain must be transmitted through the medulla oblongata; and in reflex action, whatever sensations are transmitted to the brain must in like manner pass through this singular little mass of nervous matter.

The medulla oblongata is also subdivided laterally, so that two distinct centres of sensation are found to exist in the posterior half, and two corresponding centres of motion in the anterior column. Experiments show that if one of the anterior columns of the medulla oblongata is cut away the power of motion in the opposite half of the body is lost, although in the other half motion will remain, and sensation is unimpaired; remove the other half of the anterior column, and all motion ceases. The same results to the powers of sensation are obtained by cutting away the posterior columns separately or together.

Experiments made on living animals have shown that although successive portions of the brain may be cut away until it is all removed, so long as the medulla oblongata remains uninjured the functions of breathing, deglutition, and the automatic processes of life still go on undisturbed.

A very slight wound at the centre of the medulla oblongata is sufficient to produce death, and the functions of motion and sensation are, as we have shown, absolutely dependent on its integrity and perfectness.

Many physiologists assume that the instinctive powers of animal life are centred in the cerebellum; but whilst that organ is undoubtedly essential to